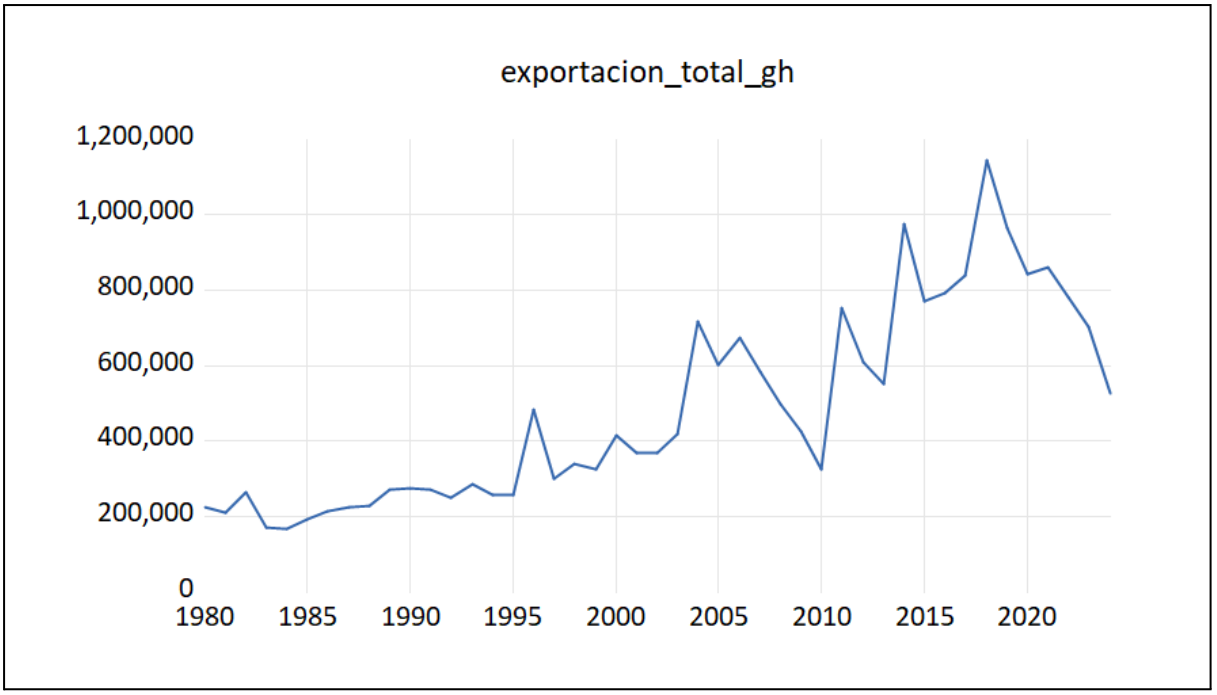


El gráfico crece muy lentamente entre 1980-2000, luego se ve una subida con una gran volatilidad, con dos grandes caídas entre 2006-2010 y 2018-2024.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: EXPORTACION TOTAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.756365	0.0287
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

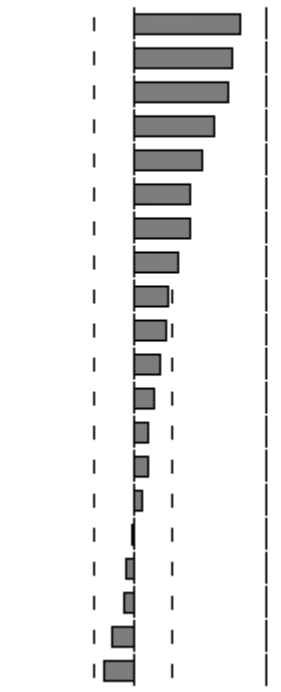
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION TOTAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:31
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION TOTAL GH(-1)	-0.582623	0.155103	-3.756365	0.0005
C	64884.70	40058.45	1.619751	0.1130
@TREND("1980")	9894.994	3182.913	3.108785	0.0034

R-squared	0.258826	Mean dependent var	6913.498
Adjusted R-squared	0.222671	S.D. dependent var	142028.8
S.E. of regression	125221.5	Akaike info criterion	26.37930
Sum squared resid	6.43E+11	Schwarz criterion	26.50095
Log likelihood	-577.3446	Hannan-Quinn criter.	26.42441
F-statistic	7.158810	Durbin-Watson stat	1.952845
Prob(F-statistic)	0.002155		

Como la tendencia determinística resulta significativo analizo el p-valor para verificar la presencia de raíz unitaria al 5% de significatividad, como el p-valor es 0.0287<0.05, Rechazo la hipótesis nula de raíz unitaria y por lo tanto analizo la serie en niveles.

Date: 10/14/26 Time: 18:32
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.843	0.843	34.164	0.000
		2	0.778	0.232	63.913	0.000
		3	0.739	0.147	91.433	0.000
		4	0.646	-0.139	112.99	0.000
		5	0.540	-0.183	128.43	0.000
		6	0.455	-0.088	139.67	0.000
		7	0.446	0.261	150.74	0.000
		8	0.360	-0.083	158.16	0.000
		9	0.277	-0.124	162.65	0.000
		10	0.258	0.029	166.69	0.000
		11	0.209	-0.026	169.41	0.000
		12	0.164	0.053	171.13	0.000
		13	0.120	-0.017	172.08	0.000
		14	0.110	-0.007	172.91	0.000
		15	0.069	-0.092	173.25	0.000
		16	-0.006	-0.138	173.25	0.000
		17	-0.053	-0.117	173.46	0.000
		18	-0.081	0.074	173.98	0.000
		19	-0.166	-0.148	176.23	0.000
		20	-0.223	-0.035	180.45	0.000

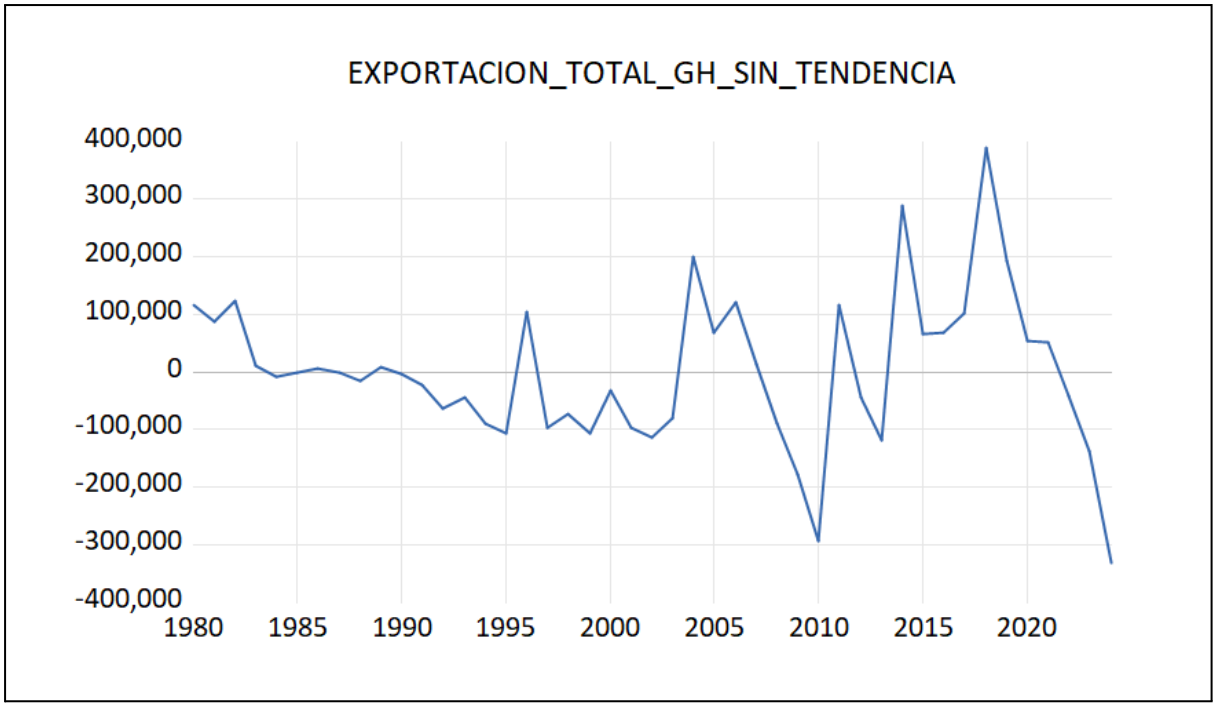
Compruebo la significatividad de la tendencia:

Dependent Variable: EXPORTACION TOTAL GH
Method: Least Squares
Date: 10/14/26 Time: 18:37
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	106537.9	39278.73	2.712355	0.0096
@TREND	17096.48	1537.488	11.11975	0.0000

R-squared	0.741972	Mean dependent var	482660.4
Adjusted R-squared	0.735972	S.D. dependent var	260679.8
S.E. of regression	133946.9	Akaike info criterion	26.49170
Sum squared resid	7.71E+11	Schwarz criterion	26.57200
Log likelihood	-594.0633	Hannan-Quinn criter.	26.52163
F-statistic	123.6488	Durbin-Watson stat	1.130227
Prob(F-statistic)	0.000000		

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 18:39
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.355	0.355	6.0601	0.014
		2	0.168	0.049	7.4566	0.024
		3	0.110	0.041	8.0679	0.045
		4	-0.058	-0.133	8.2419	0.083
		5	-0.267	-0.256	12.024	0.034
		6	-0.329	-0.195	17.903	0.006
		7	0.002	0.278	17.903	0.012
		8	-0.100	-0.094	18.477	0.018
		9	-0.205	-0.224	20.935	0.013
		10	-0.072	-0.082	21.246	0.019
		11	0.059	0.090	21.464	0.029
		12	0.011	0.048	21.473	0.044
		13	0.018	0.061	21.495	0.064
		14	0.227	0.062	25.022	0.034
		15	0.125	-0.134	26.123	0.037
		16	-0.050	-0.101	26.307	0.050
		17	-0.187	-0.191	28.938	0.035
		18	-0.104	0.015	29.782	0.040
		19	-0.207	-0.053	33.267	0.022
		20	-0.213	0.006	37.100	0.011

Se ve como cambia el correlograma con respecto al anterior, tanto la FAC como la FACP dejan de ser significativas en el segundo rezago

Dependent Variable: EXPORTACION_TOTAL_GH_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/16/26 Time: 22:20
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 17 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.562156	0.308919	1.819752	0.0759
MA(1)	-0.181383	0.361233	-0.502121	0.6182

R-squared	0.154563	Mean dependent var	-2665.806
Adjusted R-squared	0.134433	S.D. dependent var	132719.8
S.E. of regression	123476.9	Akaike info criterion	26.32989
Sum squared resid	6.40E+11	Schwarz criterion	26.41099
Log likelihood	-577.2575	Hannan-Quinn criter.	26.35996
Durbin-Watson stat	1.880576		

Inverted AR Roots	.56
Inverted MA Roots	.18

El componente MA(1) no resultó significativo, por lo que pruebo la estimación solo con el componente AR(1)

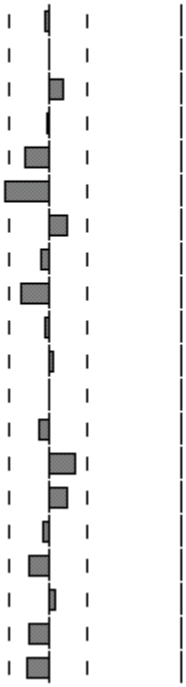
Dependent Variable: EXPORTACION_TOTAL_GH_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:44
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.413738	0.150447	2.750058	0.0087

R-squared	0.149222	Mean dependent var	-2665.806
Adjusted R-squared	0.149222	S.D. dependent var	132719.8
S.E. of regression	122417.6	Akaike info criterion	26.29073
Sum squared resid	6.44E+11	Schwarz criterion	26.33128
Log likelihood	-577.3960	Hannan-Quinn criter.	26.30577
Durbin-Watson stat	1.941199		

Inverted AR Roots	.41

Date: 10/14/26 Time: 18:44
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.030	-0.030	0.0427	
		2	0.015	0.015	0.0542	0.816
		3	0.122	0.123	0.7864	0.675
		4	-0.007	0.001	0.7886	0.852
		5	-0.178	-0.185	2.4262	0.658
		6	-0.338	-0.384	8.5242	0.130
		7	0.140	0.128	9.6028	0.142
		8	-0.057	0.047	9.7823	0.201
		9	-0.218	-0.162	12.523	0.129
		10	-0.019	-0.155	12.545	0.184
		11	0.039	-0.092	12.640	0.245
		12	0.003	-0.005	12.641	0.317
		13	-0.073	0.038	12.992	0.370
		14	0.205	0.127	15.833	0.258
		15	0.144	0.011	17.282	0.241
		16	-0.040	-0.076	17.396	0.296
		17	-0.141	-0.271	18.892	0.274
		18	0.063	-0.019	19.200	0.317
		19	-0.140	-0.083	20.795	0.290
		20	-0.164	0.007	23.062	0.235

La estimación del coeficiente del AR(1) resulto significativa, con ruido blanco, y raíz dentro del círculo unitario, sin embargo debo probar la estimación solo con el componente MA(1)

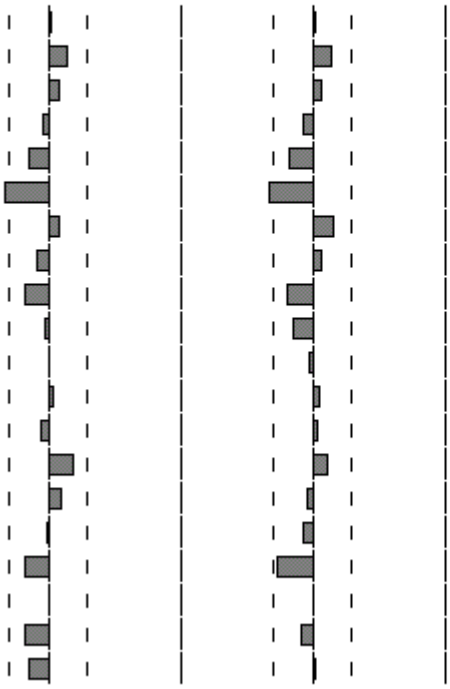
Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:42
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 14 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.365908	0.146871	2.491356	0.0166

R-squared	0.126457	Mean dependent var	1.24E-10
Adjusted R-squared	0.126457	S.D. dependent var	132416.0
S.E. of regression	123760.7	Akaike info criterion	26.31206
Sum squared resid	6.74E+11	Schwarz criterion	26.35221
Log likelihood	-591.0213	Hannan-Quinn criter.	26.32703
Durbin-Watson stat	1.798459		

Inverted MA Roots	-.37

Date: 10/14/26 Time: 18:43
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.032	0.032	0.0479	
		2	0.151	0.150	1.1684	0.280
		3	0.082	0.075	1.5090	0.470
		4	-0.042	-0.070	1.5987	0.660
		5	-0.155	-0.183	2.8715	0.580
		6	-0.336	-0.343	8.9995	0.109
		7	0.083	0.161	9.3828	0.153
		8	-0.085	0.072	9.7981	0.200
		9	-0.181	-0.197	11.731	0.164
		10	-0.030	-0.141	11.785	0.226
		11	0.015	-0.030	11.799	0.299
		12	0.038	0.056	11.890	0.372
		13	-0.052	0.042	12.071	0.440
		14	0.195	0.107	14.666	0.329
		15	0.106	-0.048	15.453	0.348
		16	-0.006	-0.079	15.456	0.419
		17	-0.177	-0.272	17.810	0.335
		18	0.010	0.002	17.817	0.400
		19	-0.182	-0.085	20.509	0.305
		20	-0.141	0.021	22.198	0.275

Como con ambos modelos llego a coeficientes significativos, ruido blanco y raíz dentro del circulo unitario, utilizo el criterio de informacion de Akaike para comparar entre modelos. Debido a que al estimar el AR(1) se resta una observacion, y el AIC requiere la misma cantidad de observaciones debo estimar el modelo MA(1) nuevamente con 44 obs.

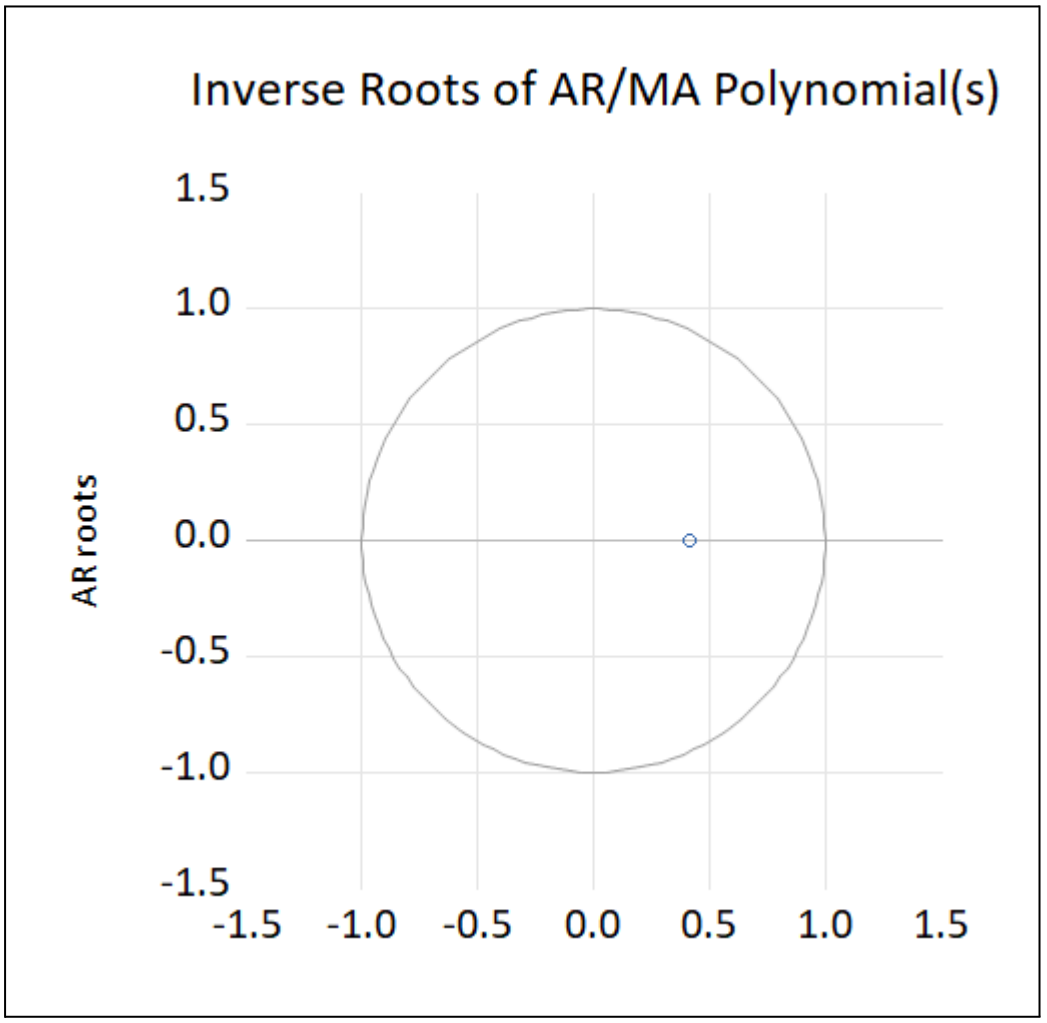
Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:45
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 15 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

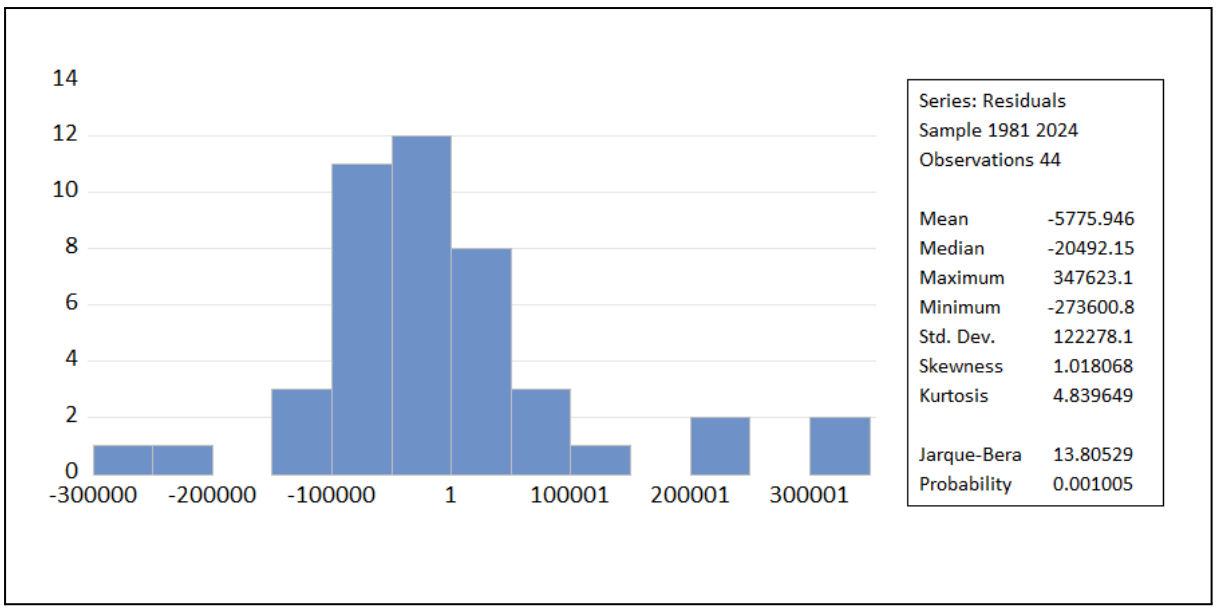
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.360762	0.149673	2.410333	0.0203

R-squared	0.122334	Mean dependent var	-2665.806
Adjusted R-squared	0.122334	S.D. dependent var	132719.8
S.E. of regression	124337.0	Akaike info criterion	26.32184
Sum squared resid	6.65E+11	Schwarz criterion	26.36239
Log likelihood	-578.0806	Hannan-Quinn criter.	26.33688
Durbin-Watson stat	1.796947		

Inverted MA Roots	-.36

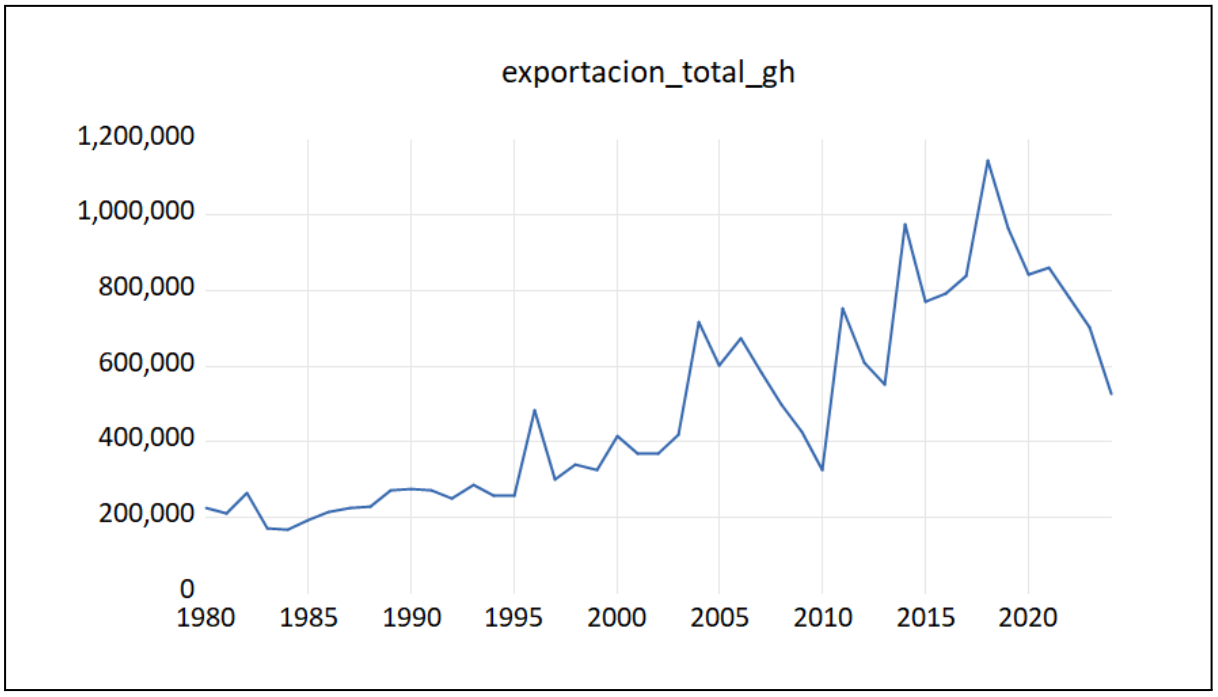
Al comparar el AIC vemos que el modelo AR(1) AIC=26.29, y en el modelo MA(1) AIC=26.32, por lo tanto elijo el modelo con el criterio mas bajo, es decir, el AR(1)





El test de Jarque-Bera del modelo AR(1) rechaza la hipótesis nula de que los residuos se comportan normalmente.

Incorporando una variable de impacto:



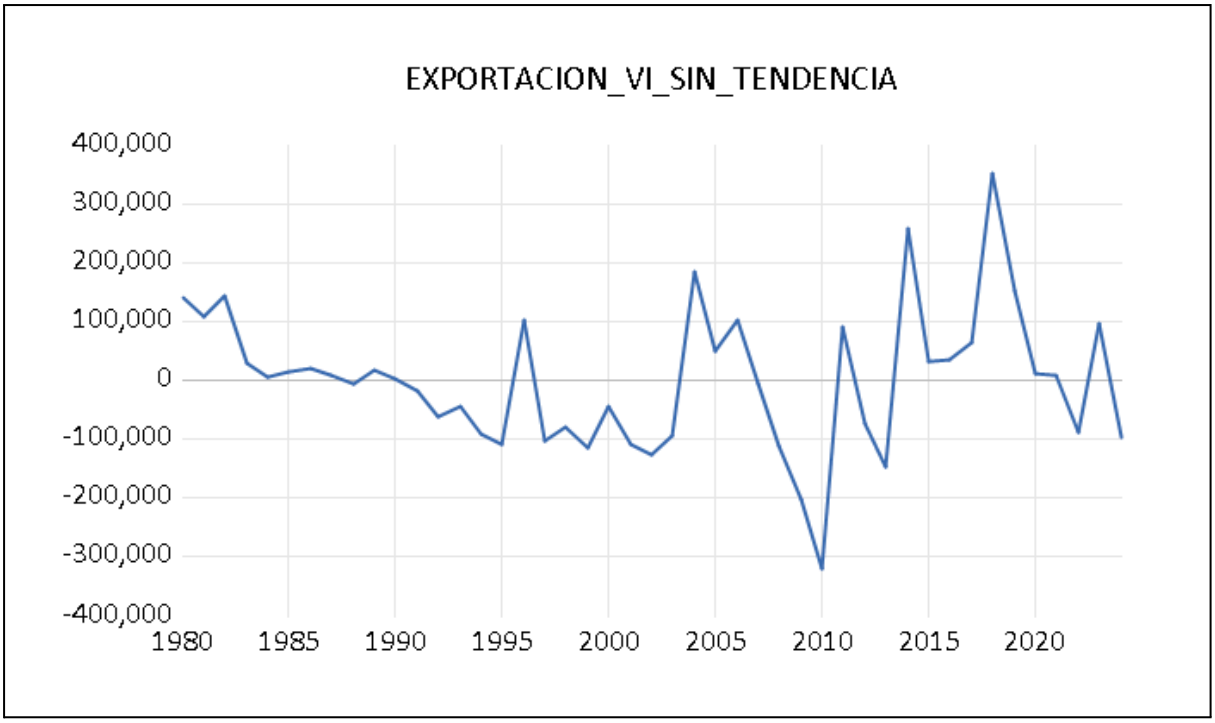
Durante los últimos años está ocurriendo una crisis respecto a la producción del cacao en África Occidental, provocada por un virus sobre la planta de cacao, que afectó a Ghana y Costa de Marfil, dos de los mayores productores de cacao a nivel mundial (60% de la oferta mundial).
Es por eso que decidí incorporar una variable de intervencion que refleje ese shock tanto en la producción como en el precio internacional, y por lo tanto en el precio del productor.
Con respecto a la variable exportacion total

Dependent Variable: EXPORTACION_TOTAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:34
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI_CRISIS	-281198.5	95567.65	-2.942403	0.0053
C	83987.63	36990.97	2.270490	0.0284
@TREND	18689.57	1516.467	12.32441	0.0000

R-squared	0.786071	Mean dependent var	482660.4
Adjusted R-squared	0.775884	S.D. dependent var	260679.8
S.E. of regression	123408.2	Akaike info criterion	26.34872
Sum squared resid	6.40E+11	Schwarz criterion	26.46917
Log likelihood	-589.8463	Hannan-Quinn criter.	26.39362
F-statistic	77.16340	Durbin-Watson stat	1.403986
Prob(F-statistic)	0.000000		

La variable de intervención que mejor refleja la crisis en África occidental es aquella que incluye a los anios 2023 y 2024.



Null Hypothesis: EXPORTACION_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

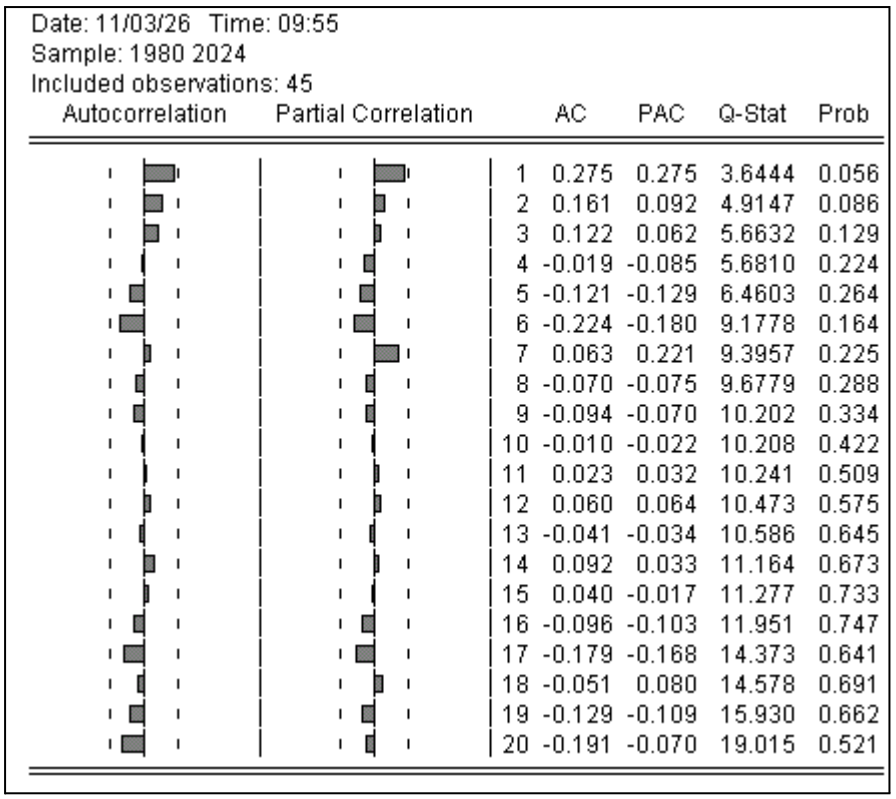
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.964373	0.0000
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

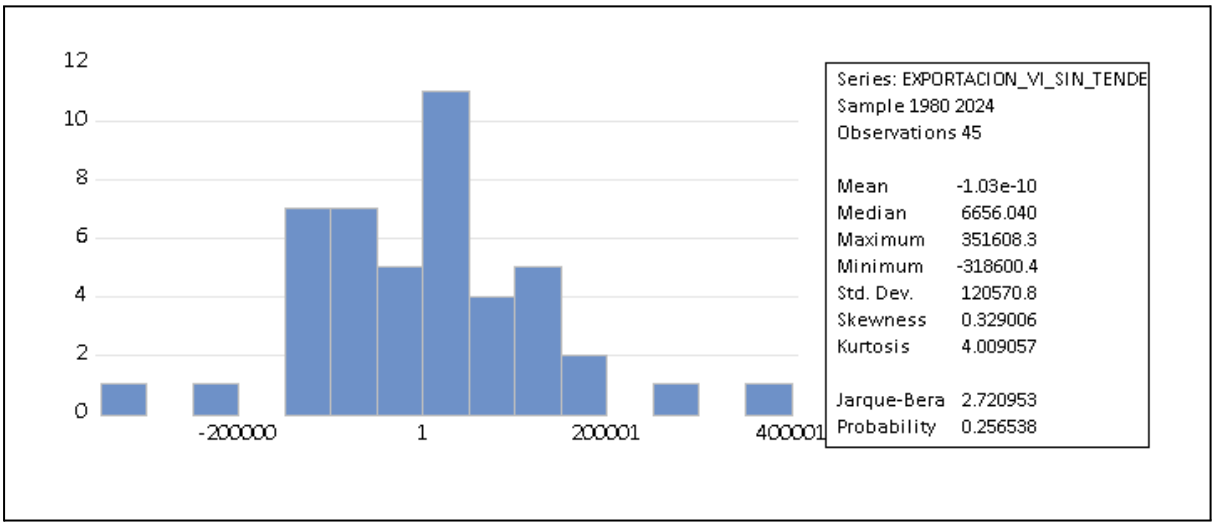
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION_VI_SIN_TENDENCIA)
Method: Least Squares
Date: 11/02/26 Time: 12:36
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION_VI_SIN_TENDENCIA(-1)	-0.720531	0.145140	-4.964373	0.0000

R-squared	0.363424	Mean dependent var	-5385.196
Adjusted R-squared	0.363424	S.D. dependent var	144413.2
S.E. of regression	115221.1	Akaike info criterion	26.16956
Sum squared resid	5.71E+11	Schwarz criterion	26.21011
Log likelihood	-574.7303	Hannan-Quinn criter.	26.18460
Durbin-Watson stat	2.048261		



	AC	PAC	Q-Stat	Prob
1	0.275	0.275	3.6444	0.056
2	0.161	0.092	4.9147	0.086
3	0.122	0.062	5.6632	0.129
4	-0.019	-0.085	5.6810	0.224
5	-0.121	-0.129	6.4603	0.264
6	-0.224	-0.180	9.1778	0.164
7	0.063	0.221	9.3957	0.225
8	-0.070	-0.075	9.6779	0.288
9	-0.094	-0.070	10.202	0.334
10	-0.010	-0.022	10.208	0.422
11	0.023	0.032	10.241	0.509
12	0.060	0.064	10.473	0.575
13	-0.041	-0.034	10.586	0.645
14	0.092	0.033	11.164	0.673
15	0.040	-0.017	11.277	0.733
16	-0.096	-0.103	11.951	0.747
17	-0.179	-0.168	14.373	0.641
18	-0.051	0.080	14.578	0.691
19	-0.129	-0.109	15.930	0.662
20	-0.191	-0.070	19.015	0.521



Al ver el correlograma luego de eliminar la tendencia, el intercepto y el efecto de la crisis de los anios 23/24 arroja directamente ruido blanco, por lo que no hay ningun componente AR/MA que incluir en el modelo.
Además el comportamiento de la nueva serie no rechaza el test de normalidad.